

Approximate Fourier Series of the Magnetic Field Induced by a Single Picture-Frame Coil

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In this document we will calculate the fourier series in toroidal angle, ϕ , for a single picture-frame coil. The picture-frame coil is made up of two infinitely long vertical wires which extend from $z \rightarrow -\infty$ to $z \rightarrow \infty$ and two horizontal wires which extend from $x = 0$ to $x \rightarrow \infty$. The current in the vertical wires is I_0 . Note that we use infinite wires to simplify the calculations. We will later show numerically that our values agree well with the exact values obtained numerically. We calculate for a single picture-frame coil, however, it's trivial to extend the results to N picture-frame coils evenly space around a Tokamak.

1 Fourier series of vertical wire at $x = R_0, y = 0$

The magnetic field induced by an infinite vertical wire at $x = R_0$ and $y = 0$ is given by

$$\begin{aligned}\mathbf{B}_v(R, \phi, z; R_0) &= \frac{\mu_0 I_0}{2\pi} \frac{-y\hat{\mathbf{x}} + (x - R_0)\hat{\mathbf{y}}}{(x - R_0)^2 + y^2} \\ &= \frac{\mu_0 I_0}{2\pi R} \frac{-R_0 y \hat{\mathbf{R}} + (R^2 - R_0 x) \hat{\phi}}{(x - R_0)^2 + y^2} \\ &= \frac{\mu_0 I_0}{2\pi R} \left\{ \text{Im} \left[\frac{1}{1 - (R_0/R) \exp(-i\phi)} \right] \hat{\mathbf{R}} + \text{Re} \left[\frac{1}{1 - (R_0/R) \exp(-i\phi)} \right] \hat{\phi} \right\} \\ &= -\frac{\mu_0 I_0}{2\pi R} \left\{ \text{Im} \left[\frac{(R/R_0) \exp(i\phi)}{1 - (R/R_0) \exp(i\phi)} \right] \hat{\mathbf{R}} + \text{Re} \left[\frac{(R/R_0) \exp(i\phi)}{1 - (R/R_0) \exp(i\phi)} \right] \hat{\phi} \right\}.\end{aligned}$$

where

$$\begin{aligned}R &= x^2 + y^2, \\ x &= R \cos \phi, \\ y &= R \sin \phi, \\ \hat{\mathbf{R}} &= \frac{1}{R} [x\hat{\mathbf{x}} + y\hat{\mathbf{y}}], \\ \hat{\phi} &= \frac{1}{R} [-y\hat{\mathbf{x}} + x\hat{\mathbf{y}}].\end{aligned}$$

Hence, using the formula for a geometric series,

$$\mathbf{B}_v(R, \phi, z; R_0) = \frac{\mu_0 I_0}{2\pi R} \sum_{n=0}^{\infty} \left\{ - \left[\left(\frac{R_0}{R} \right)^n \sin(n\phi) \right] \hat{\mathbf{R}} + \left[\left(\frac{R_0}{R} \right)^n \cos(n\phi) \right] \hat{\phi} \right\}.$$

for $R > R_0$ and

$$\mathbf{B}_v(R, \phi, z; R_0) = -\frac{\mu_0 I_0}{2\pi R} \sum_{n=1}^{\infty} \left\{ \left[\left(\frac{R}{R_0} \right)^n \sin(n\phi) \right] \hat{\mathbf{R}} + \left[\left(\frac{R}{R_0} \right)^n \cos(n\phi) \right] \hat{\phi} \right\},$$

for $R < R_0$.

2 Fourier series of horizontal wire at $y = z = 0$

The magnetic field induced by a vertical wire which extends from $z = z_{min}$ to $z = z_{max}$ can be calculated using the Biot-Savart law to give

$$\mathbf{B}'_v(R, \phi, z) = \frac{\mu_0 I_0}{4\pi R} \left[\frac{z_{max} - z}{\sqrt{R^2 + (z_{max} - z)^2}} + \frac{z - z_{min}}{\sqrt{R^2 + (z - z_{min})^2}} \right] \hat{\phi}.$$

Hence, the magnetic field induced by a horizontal wire at $y = z = 0$ and extends from $x = 0$ to $x \rightarrow \infty$ is given by

$$\begin{aligned} \mathbf{B}_h(x, y, z) &= \frac{\mu_0 I_0}{4\pi} \frac{-z\hat{\mathbf{y}} + y\hat{\mathbf{z}}}{y^2 + z^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + z^2}} \right] \\ &= \frac{\mu_0 I_0}{4\pi} \frac{-z \sin \phi \hat{\mathbf{R}} - z \cos \phi \hat{\phi} + R \sin \phi \hat{\mathbf{z}}}{R^2 \sin^2 \phi + z^2} \left[1 + \frac{R \cos \phi}{\sqrt{R^2 + z^2}} \right]. \end{aligned}$$

We can show that

$$\begin{aligned} B_{R,h}(R, \phi, z) &= -\frac{\mu_0 I_0}{2\pi} \frac{1}{R} \frac{z}{\sqrt{R^2 + z^2}} \sum_{n=1}^{\infty} \left(\frac{\sqrt{R^2 + z^2} - |z|}{R} \right)^n \sin(n\phi), \\ B_{\phi,h}(R, \phi, z) &= \frac{\mu_0 I_0}{2\pi} \frac{1}{R} \left\{ \frac{z}{2\sqrt{R^2 + z^2}} - \frac{z}{2|z|} - \frac{z}{|z|} \sum_{n=1}^{\infty} \left(\frac{\sqrt{R^2 + z^2} - |z|}{R} \right)^n \cos(n\phi) \right\}, \\ B_{z,h}(R, \phi, z) &= \frac{\mu_0 I_0}{2\pi} \frac{1}{R} \frac{R}{\sqrt{R^2 + z^2}} \sum_{n=1}^{\infty} \left(\frac{\sqrt{R^2 + z^2} - |z|}{R} \right)^n \sin(n\phi). \end{aligned}$$

To prove the above is quite involved, however, by going to the following Desmos link you can see for yourself that the above is almost certainly true: <https://www.desmos.com/calculator/xy18dvtot4>.

3 Fourier series of a single picture-frame coil

Combining results from the previous section, we can write the magnetic field for a single picture-frame coil as

$$\mathbf{B}(R, \phi, z) = \mathbf{B}_v(R, \phi, z; R_{inner}) - \mathbf{B}_v(R, \phi, z; R_{outer}) + \mathbf{B}_h(R, \phi, z - h) - \mathbf{B}_h(R, \phi, z + h),$$

where R_{inner} and R_{outer} are the inner and outer radii of the picture-frame coil, respectively, and h is the half the height of the picture-frame coil. We can write each component as

$$\begin{aligned} B_R(R, \phi, z) &= \frac{\mu_0 I_0}{2\pi R} \sum_{n=1}^{\infty} \left[\left(\frac{R}{R_{outer}} \right)^n - \left(\frac{R_{inner}}{R} \right)^n + \right. \\ &\quad \frac{h - z}{\sqrt{R^2 + (z - h)^2}} \left(\frac{\sqrt{R^2 + (z - h)^2} - (h - z)}{R} \right)^n + \\ &\quad \left. \frac{z + h}{\sqrt{R^2 + (z + h)^2}} \left(\frac{\sqrt{R^2 + (z + h)^2} - (z + h)}{R} \right)^n \right] \sin(n\phi), \end{aligned}$$

$$\begin{aligned} B_{\phi}(R, \phi, z) &= \frac{\mu_0 I_0}{2\pi R} \left\{ 1 + \sum_{n=1}^{\infty} \left[\left(\frac{R_{inner}}{R} \right)^n + \left(\frac{R}{R_{outer}} \right)^n + \right. \right. \\ &\quad \left(\frac{\sqrt{R^2 + (z - h)^2} - (h - z)}{R} \right)^n + \\ &\quad \left. \left. \left(\frac{\sqrt{R^2 + (z + h)^2} - (z + h)}{R} \right)^n \right] \cos(n\phi) \right\}, \end{aligned}$$

$$B_z(R, \phi, z) = \frac{\mu_0 I_0}{2\pi R} \sum_{n=1}^{\infty} \left[\frac{R}{\sqrt{R^2 + (z-h)^2}} \left(\frac{\sqrt{R^2 + (z-h)^2} - (h-z)}{R} \right)^n - \frac{R}{\sqrt{R^2 + (z+h)^2}} \left(\frac{\sqrt{R^2 + (z+h)^2} - (z+h)}{R} \right)^n \right] \sin(n\phi),$$

for $R_{inner} < R < R_{outer}$, $-h < z < h$. Note that the $n = 0$ term we set to $\frac{\mu_0 I_0}{2\pi} \hat{\phi}$, despite the formula from the previous section suggesting otherwise. This is because we know the $n = 0$ term from Ampere's law and the difference is caused by our use of infinite wires instead of finite wires.

4 Fourier series of finite length horizontal wire at $y = z = 0$

and extends from $x = R_{min}$ to $x = R_{max}$ is given by

$$\begin{aligned} \mathbf{B}'_v(R, \phi, z) &= \frac{\mu_0 I_0}{4\pi R} \left[\frac{z_{max} - z}{\sqrt{R^2 + (z_{max} - z)^2}} + \frac{z - z_{min}}{\sqrt{R^2 + (z - z_{min})^2}} \right] \hat{\phi}. \\ \mathbf{B}_h &= \frac{\mu_0 I_0}{4\pi} \frac{-z\hat{\mathbf{y}} + y\hat{\mathbf{z}}}{y^2 + z^2} \left[\frac{R_{max} - x}{\sqrt{(R_{max} - x)^2 + y^2 + z^2}} + \frac{x - R_{min}}{\sqrt{(x - R_{min})^2 + y^2 + z^2}} \right] \\ &= \frac{\mu_0 I_0}{4\pi R_{max}} \frac{-z'\hat{\mathbf{y}} + y'\hat{\mathbf{z}}}{y'^2 + z'^2} \left[\frac{1 - x'}{\sqrt{(1 - x')^2 + y'^2 + z'^2}} + \frac{x'' - R''_{min}}{\sqrt{(x'' - R''_{min})^2 + y''^2 + z''^2}} \right] \\ &= \frac{\mu_0 I_0}{4\pi R_{max}} \frac{-z' \sin \phi \hat{\mathbf{R}} - z' \cos \phi \hat{\phi} + R' \sin \phi \hat{\mathbf{z}}}{R'^2 \sin^2 \phi + z'^2} \left[\frac{1 - R' \cos(\phi)}{\sqrt{(1 - R' \cos \phi)^2 + R'^2 \sin^2 \phi + z'^2}} + \frac{\cos \phi - R''_{min}}{\sqrt{(\cos^2 \phi - R''_{min})^2 + \sin^2 \phi + z''^2}} \right] \\ &= \frac{\mu_0 I_0}{4\pi R_{max}} \frac{-z' \sin \phi \hat{\mathbf{R}} - z' \cos \phi \hat{\phi} + R' \sin \phi \hat{\mathbf{z}}}{R'^2 \sin^2 \phi + z'^2} \left[\frac{1 - R' \cos(\phi)}{\sqrt{1 + R'^2 + z'^2 - 2R' \cos(x)}} + \frac{\cos \phi - R''_{min}}{\sqrt{1 + R''_{min}^2 + z''^2 - 2R''_{min} \cos(x)}} \right]. \end{aligned}$$

where $'$ denotes divided by R_{max} , e.g. $x' = x/R_{max}$, $R' = R/R_{max}$, etc and $''$ denotes divided by R , e.g. $x'' = x/R$, $R''_{min} = R_{min}/R$, etc.

5 Claim

Our claim is that

$$\begin{aligned} B_{R,n}(R, \phi, z) &\leq \left[\left(\frac{R}{R_{outer}} \right)^n - \left(\frac{R_{inner}}{R} \right)^n + \frac{h-z}{\sqrt{R^2 + (z-h)^2}} \left(\frac{\sqrt{R^2 + (z-h)^2} - (h-z)}{R} \right)^n + \frac{z+h}{\sqrt{R^2 + (z+h)^2}} \left(\frac{\sqrt{R^2 + (z+h)^2} - (z+h)}{R} \right)^n \right] \end{aligned}$$

and

$$\begin{aligned} B_{R,n}(R, \phi, z) &\geq \max \left[\left(\frac{R}{R_{outer}} \right)^n - \left(\frac{R_{inner}}{R} \right)^n, \frac{h-z}{\sqrt{R^2 + (z-h)^2}} \left(\frac{\sqrt{R^2 + (z-h)^2} - (h-z)}{R} \right)^n + \frac{z+h}{\sqrt{R^2 + (z+h)^2}} \left(\frac{\sqrt{R^2 + (z+h)^2} - (z+h)}{R} \right)^n \right] \end{aligned}$$